

1) Image Enhancement in the Spatial Domain

Image Enhancement is the process of improving the quality, contrast, brightness and details of an image.

The basic process is:

Input Image $\rightarrow$ pixel processing $\rightarrow$ Enhanced Image

Main techniques:

1) Gray level Transformations

These transformations change the intensity values of pixels to improve brightness and contrast.

Examples:

image negative, logarithmic transformation

2) Histogram processing

A histogram shows the distribution of intensity values in an image.

3) Spatial Filtering

Spatial filters process a pixel using its neighboring pixels.

$\Rightarrow$ Smoothing filters such as mean and median filters reduce noise

$\Rightarrow$ Sharpening filters such as Laplacian and Sobel filters

2) Image Smoothing and image Sharpening are important image enhancement techniques, but opposite purposes.

Smoothing	Sharpening
Reduce noise	Enhances edges and details
Makes the image smoother	Makes the image clearer
Suppresses high-frequency components	Enhances high-frequency components
May blur edges	Makes edges more prominent
Mean, Gaussian, Median filters	Laplacian, Sobel filters

Spatial smoothing filters:

Mean filters: Calculates the average of neighbouring pixels

Gaussian filters: Uses weighted neighboring pixels to produce smooth results

Median filters: Replaces the pixel with the median value and is especially useful for salt and pepper

Sharpening filters: sharpening filters emphasize sudden changes in intensity

Filters: Uses second-order derivatives to enhance fine details horizontal and vertical

Filters: Detects edges.

Applications:

Smoothing is applied when the image contains noise, while sharpening is applied when edges, boundaries or fine details need to be clearly visible.

3) Frequency-domain enhancement processes an image based on its frequency components. The image is first converted from the spatial domain to the frequency domain using the Fourier transform.

Input Image

↓
Fourier Transform

↓
Apply Filter

↓
Inverse Fourier Transform

↓
Enhanced Image

The two important filters are low-pass filters (LPF) and High Pass Filter (HPF)

Low pass Filter

A low pass filter allows low frequency components to pass and reduce high-frequency components.

Effects:

- ⇒ Smooths the image
- ⇒ Reduce noise
- ⇒ Removes small unwanted details

Applications:

Noise reduction, image smoothing and preprocessing

High pass Filter

A high-pass filter allows high-frequency components to pass and reduce low-frequency components.

Effects:

- ⇒ Enhance edges
- ⇒ Improve fine details
- ⇒ Sharpen the image

Applications:

Edge enhancement, sharpening, object recognition and defect detection.

Comparison

LPF → Smoother image → Noise reduction

HPF → Sharper image → Edge/details enhancement

4)

Image Segmentation is the process of dividing an image into different meaningful regions or objects.

Types of Segmentation

high-contrast

Isolated pixels that are significantly different from their surroundings.

Detection

Detects straight lines in different directions such as Horizontal, vertical and diagonal.

3. Edge Detection:

Detects sudden change in Intensity and identifies object boundaries. Operators such as Sobel, Prewitt and Canny are commonly used.

4. Thresholding:

Separates an object from the background based on pixel intensity.

$$g(x,y) = \begin{cases} 1, & f(x,y) > T \\ 0, & f(x,y) \leq T \end{cases}$$

5. Region based Segmentation

Groups neighbouring pixels having similar properties such as intensity or color.

Examples

region growing and region splitting

Suitable Technique:

for object and defect detection, edge detection and thresholding are most suitable.

edges are important features in an image because they indicate a sudden change in intensity.

1. Edge Detection

Edge detection identifies pixels where there is a significant change in intensity. Common techniques include

- ⇒ Sobel
- ⇒ Prewitt
- ⇒ Roberts
- ⇒ Canny

2. Edge linking

Edge detection may produce broken or disconnect edge segments.

Before: — —

✓ After: — — — —

3. Boundary Detection

Boundary detection identifies the complete outline of an object.



Importance

Image → Edge Detection → Edge linking → Boundary Detection → object Recognition